THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2026** – JUNIOR FINAL

Timings: 10am to 1pm

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Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.

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1. ΔABC is a right-angled triangle, ∠ACB = 90°. The sides are of integral length. If the product of the legs is equal to the sum of all sides, then find the sides of the triangle. How many such triangles are there?

2. n is a natural number and $n \leq 2026$. If n divides $20^n + 17k$, for some natural number k, then find the number of such n.

3. In triangle ABC, the incircle, with center I, touches the side BC at point D. Line DI meets AC at P. The tangent line from P to the incircle (different from AC) intersects AB at Q. If line QI and BC intersect at point R, prove that quadrilateral AIRC is a cyclic.

4. Given a sequence: $a_1, a_2, a_3, \ldots,$ where $a_n = a_{n-1} - a_{n-2}$, for $n \in \mathbb{N}$ and $n \geq 3$.
   Also, $S_n = a_1 + a_2 + a_3 + \ldots + a_{n-1} + a_n$. If $S_{2003} = 2026$ and $S_{2026} = 2003$, then find S_{2030}.

5. Find all possible values of natural numbers m and n such that
   $$12mn - 175 = 105m + 140n.$$

6. Find all natural numbers x, y, z such that
   $$\sqrt{\frac{2026}{x+y}} + \sqrt{\frac{2026}{y+z}} + \sqrt{\frac{2026}{z+x}}$$
   is also a natural number.

7. How many ways can you mark 8 squares of an 8 × 8 chessboard so that no two marked squares are in the same row or column, and none of the four corner squares is marked? (Rotations and reflections are considered different.)

8. In a triangle ABC we have ∠ACB = ∠BAC + 90°. The point D is taken on the extended BC such that AC = AD. A point E is taken such that points A and E are on the opposite sides of side BC and ∠EBC = ∠BAC, ∠EDC = $\frac{1}{2}$∠BAC.
   Prove that ∠CED = ∠ABC.

------x------x------x------